

Education	University of Virginia <i>Doctor of Philosophy, Data Science</i> • Advisor: Dr. Stephen Baek	Aug 2025 – May 2030 (expected)
	Kean University <i>Master's, Computational Science & Engineering (transferred to PhD)</i> • Advisor: Dr. David Joiner	Jan 2025 – June 2025
	Kean University <i>Bachelor of Science, Computational Science & Engineering</i> • Minor: Computer Science & Applied Mathematics	Sep 2021 – Dec 2024
Work Experience	RCSB Protein Data Bank Scientific Software Developer Intern • Developed and published the `rcsb-api` Python package on GitHub and PyPI, streamlining programmatic access to RCSB Protein Data Bank structural biology data through automated API querying and retrieval. [A1]	<i>New Jersey, USA</i> May 2024 – Aug 2024
Research Summary	I study physics-aware machine learning, computational fluid dynamics, and generative design for complex engineering systems, with applications in aerodynamic vehicle design and energetic materials. My research focuses on integrating simulation, deep learning, and geometry optimization to build efficient surrogate models and accelerate physics-based design exploration.	
Research Interests	Physics-aware deep learning, computational fluid dynamics, generative design, aerodynamic optimization, vehicle performance and AI for engineering systems	
Publications & Posters	[A1] Piehl, D. W., Vallat, B., Truong, I., Morsy, H., et al. “rcsb-api: Python Toolkit for Streamlining Access to RCSB Protein Data Bank APIs.” <i>Journal of Molecular Biology</i>, 437(15), 168970, 2025. https://doi.org/10.1016/j.jmb.2025.168970 [PyPi] [code] [A2] Morsy, H., Toone, E., Day, C., Wang, Z., Thomas, M., & Joiner, D. “HPC-ED: Testing Automated Agents to Assess the Quality of Training Resource Metadata.” <i>Journal of Computational Science Education</i>, 17(1), 34–41, 2026. https://doi.org/10.22369/issn.2153-4136/17/1/5 [A3] Choi, J. B., Nguyen, Y. T., Cheng, X., Morsy, H., Udaykumar, H. S., & Baek, S. “Towards Foundation Models for Shock Compression of Solid Reactive Materials.” <i>55th International Annual Conference of the Fraunhofer ICT</i>, Karlsruhe, Germany, June 23–26, 2026. [A4] Morsy, H., Gray, Z., Williams, C., & Baek, S. “Beyond Idealized Voids: Linking Pore Morphology to Hotspot Sensitivity via a Statistical Shape Model.” <i>Women in Machine Learning Workshop (WiML) at NeurIPS 2026</i>, Poster, 2026	

Current Projects	Physics-Aware Deep Learning for Aerodynamic Design (PhD Dissertation)	
	Developing a physics-aware machine learning framework that integrates parametric vehicle geometry, CFD simulation, and neural surrogate modeling to accelerate aerodynamic design exploration and optimization under Formula One technical constraints	
	Beyond Idealized Voids: Linking Pore Morphology to Hotspot Sensitivity	
	Developed a Statistical Shape Model (SSM) to construct an interpretable latent design space for generating realistic microstructures and quantifying how morphological variations influence shock-induced hotspot sensitivity in energetic materials using LatentPARC.	
Extracurriculars	Aerodynamics & Suspension Team Member, <i>Virginia Motorsports</i>	2026
	Egypt National Team Taekwondo Athlete	2018-2024
	Squash Club, <i>Vice President</i>	2026
	Association for Computing Machinery (ACM), <i>President, Kean University</i>	2024
	Taekwondo Team Coach, <i>University of Virginia</i>	2026
Honors & Awards	School of Data Science PhD Showcase Best Oral Presentation, <i>University of Virginia</i>	2026
	Provost's Fellowships, <i>University of Virginia</i>	2025-2027
	Best Oral Presentation, <i>Rutgers RISE REU Symposium, New Brunswick, NJ</i>	2024
	Poster Presentation Award (Sci. & Eng.), <i>ABRCMS, Pittsburgh, PA</i>	2024
	Research Award, <i>Kean University</i>	2024
Teaching	Generation Google Scholar	2023
	Graduate Teaching Assistant, University of Virginia	
	• Data Eng II: Big Data Systems: Master's Level	08/2026
	• Machine Learning in Systems and Network Security: Master's Level	05/2026
	• Introduction to Deep Learning: Undergraduate Level	01/2026
Skills	• Computational Probability: Undergraduate Level	09/2025
	Student Assistant, Kean University	
	• Tutored Computational Physics, Python and Calculus II	2022-2024
	Programming Language: Python, C# Java, SQL, MATLAB, JavaScript, PySpark, R	
	Engineering & Scientific Software: OpenFOAM, ParaView, Blender, FreeCAD, Git, Azure DevOps, PowerShell, ANSYS Fluent, Fusion360, Unity	
	Machine Learning: PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, NetworkX, igraph, RustworkX	
	Engineering Methods: CFD, numerical simulation, surrogate modeling, 3D CAD, parametric geometry, data visualization, automation, HPC	
	Languages: English (Proficient), French (Proficient), Arabic (Native)	